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Menopause and hormone replacement therapy (HRT)

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objectives

- Know the definition of menopause.
- Understand physiological and non-physiological menopause.
- Understand the effect of menopause on women.
- Understand the modifiable and non-modifiable aspects of menopausal health.
- Explain the main forms of treatment of the menopause.
- Know the side-effects and the relative and absolute contraindications of hormonal replacement therapy (HRT).
- Describe the benefits of hormonal and non-hormonal HRT

The menopause is defined as the woman's final menstrual period and the accepted confirmation of this is made retrospectively after 1 year of amenorrhoea.

The cause of the menopause is cessation of regular ovarian function.

There is great heterogeneity in the experiences of different women as they go through the menopausal transition and several descriptive phrases exist.

Descriptive terms for menopause :

- Menopause: the last menstrual period (LMP).
- Perimenopause: time of life from the onset of ovarian dysfunction until 1 year after the last period and the diagnosis of menopause is made. This time is also known as the climacteric.
- Postmenopause: all women who have been 1 year since their last period are deemed postmenopausal.
- The 'change': a colloquial description of perimenopause and postmenopause

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Physiological menopause

- The median age of the menopause is between 51 and 52 years, with 95% of women attaining menopause between the age of 45 and 55 years.
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Endocrine changes

reproductive function is maintained by a subtle interplay of the

- hypothalamic production of gonadotrophin-releasing hormone (GnRH),
- the pituitary hormones luteinizing hormone (LH) and follicle-stimulating hormone (FSH),
- the ovarian peptide hormone inhibin B and the steroid hormones oestrogen, progesterone and testosterone.

These hormones not only change during the menstrual cycle but also throughout a woman’s reproductive life, with their production changing at differing times and rates according to the age of the woman.

Inhibin B is a peptide produced by follicles within the ovary it inhibits the release of GnRH, so as the number of follicles decline the production of inhibin decreases.

In the perimenopausal years small declines in inhibin drive an overall increase in the pulsatility of GnRH secretion and overall serum FSH and LH levels, which results in an increased drive to the remaining follicles in an attempt to maintain follicle production and oestrogen levels.

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Androgenic hormone production comes from ovaries, peripheral adipose tissue and the adrenal glands, with the ovaries producing approximately 30–50% of total circulating levels.

- A decline in ovarian testosterone and other androgens accompanies the process of ageing in women.

Overall androgen concentrations in a woman in her 20s are approximately double those at 40 years old and then slowly decline over the rest of her life, with very low levels maintained from about 70 years.

production and changes around the menopausal years

Hormones	Perimenopause	Early postmenopause	Late postmenopause and elderly
GnRH	Increased pulsatility	Progressive decrease in pulsatility	Reduction in overall levels
LH & FSH	Increased	Increased	Progressive decline
Oestrogen	Slight declines	Rapid decline in levels	Sustained very low levels
Progesterone	Moderate falls	Unpredictable	Undetectable
Inhibin	Slight decline	Significant decline	Undetectable
Testosterone	Progressive decline	Progressive decline	Sustained low levels

FSH, follicle-stimulating hormone; GnRH, gonadotrophin-releasing hormone; LH, luteinizing hormone.

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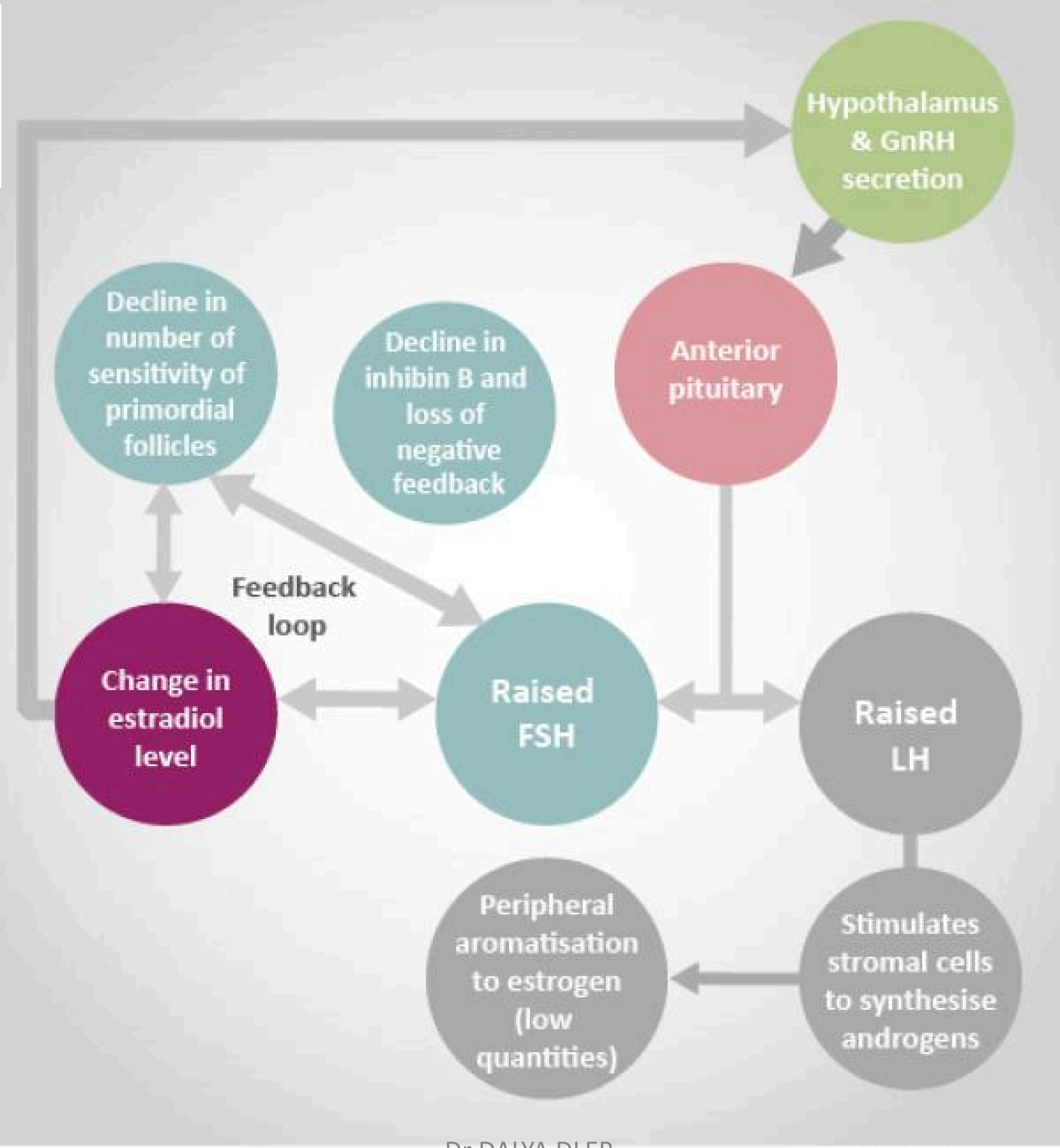

The diagnosis of menopause is a largely clinical diagnosis that is made according to symptoms experienced, such as menstrual irregularities and amenorrhoea, and oestrogen deficiency symptoms, such as vasomotor symptoms.

- The use of serum endocrine tests such as hormone levels are of little value in the perimenopausal years.
- An elevated serum FSH in association with a low serum oestradiol may be suggestive of menopause, but as this combination of levels can occur during a normal menstrual cycle this test can therefore be misleading.

In all women in whom a diagnosis of menopause is being considered, the possibility of pregnancy should also be considered.

Whether there is a „male menopause‟ or „andropause‟ is debatable. Spermatogenesis continues throughout life but incidence of loss of libido, impotence and inability to reach orgasm increase with age.

Testosterone levels may decrease but without rise in LH (or FSH) suggesting a change in the feedback control mechanism.

How women are affected by the menopause

The hormonal changes during and after the menopause have radical changes on the woman.
The timing of when each system is affected not only varies dramatically between women, but also the degree of how the changes influence each woman is remarkably unpredictable.

While most effects of the menopause have long-term implications, the effects of menopause are commonly categorized as having an early onset or an onset in the medium to long term

Effects of the menopause by time of onset

Immediate (0–5 years)	Vasomotor symptoms, (e.g. hot flushes, night sweats) Psychological symptoms (e.g. labile mood, anxiety, tearfulness) Loss of concentration, poor memory Joint aches and pains Dry and itchy skin Hair changes Decreased sexual desire
Intermediate (3–10 years)	Vaginal dryness, soreness Dyspareunia Urgency of urine Recurrent urinary tract infections Urogenital prolapse
Long term (>10 years)	Osteoporosis Cardiovascular disease Dementia

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Central nervous system

Vasomotor symptoms :

- often appear during the perimenopausal years. The colloquial term applied to vasomotor symptoms is ‘hot flush’, and when a hot flush occurs at night it is termed ‘night sweat’.

- hot flushes tend to appear unpredictably, additional triggers include alcohol, caffeine and smoking. Women with a high body mass index (BMI) tend to get worse vasomotor symptoms.

Psychological symptoms

While there is little evidence to support a direct effect of the menopause as a cause for depression, it is clear that menopause is associated with low mood, irritability, lack of energy, tiredness and impaired quality of life from the early perimenopausal period.

Cognitive function

At present there is no clear evidence that menopause is associated with an acceleration of the onset or incidence of dementia.

Endometrial effects

The initial irregular or scanty vaginal bleeding is due to the reduction in oestrogenic endometrial stimulation with failing ovarian function.

- Episodic and infrequent ovulation with fluctuations in oestrogen levels leads to unpredictable progestogenic levels, which usually has the effect of inadequate regular endometrial shedding.

The urogenital tract and vulvovaginal atrophy

Once oestrogen levels start to fall in the perimenopausal years, many women may become aware of vaginal dryness, irritation, burning, soreness and dyspareunia.

incontinence and prolapse will worsen during menopause.

Osteoporosis

is defined as ‘a skeletal disorder characterized by compromised bone strength predisposing to an increased risk of fracture’.

It is more frequent in women than men with an approximate ratio of 4:1. this happen due to increase of bone resorption by osteoclast with decline osteoblast laying down bone.

Risk factors for osteoporosis

- Family history of osteoporosis or hip fracture.
- Smoking.
- Alcoholism.
- Long-term steroid use.
- premature ovarian insufficiency and hypogonadism.
- Medical treatment of gynaecological conditions with induced menopause (like GnRH analogues).
- Disorders of thyroid and parathyroid metabolism.
- Immobility.
- Disorders of gut absorption, malnutrition, liver disease.

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Cardiovascular system:

- Oestrogen has a supportive effect on the vessel wall that favours vasodilatation and prevents atherogenesis; effects that are reduced after the menopause.

Assessment of the menopausal patient

- There is rarely a need for investigations to confirm menopause.
- While a serum FSH level more than 30 IU/l is highly suspicious of menopause, the diagnosis can be confidently made in most women based on history alone;
- The key features being oligo/amenorrhoea and the typical symptoms of vasomotor symptoms, joint aches and minor cognitive changes.

Managing menopausal symptoms

- For some women, symptoms of the menopause are mild and of short duration and may not require hormonal treatment.
- Other women, who find these symptoms distressing, may require treatment with hormone replacement therapy (HRT).
- **Non-hormonal alternatives** may be suitable for women not wishing to take hormonal therapy, or those with a contraindication to HRT.

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Management

- . **Lifestyle measures**
- Lifestyle modifications such as exercise , lighter clothing, sleeping in a cooler room, and reducing stress may be sufficient to manage hot flushes for many women.
- Avoidance of possible triggers such as spicy foods, caffeine, smoking, and alcohol may help.

Alternative and complementary treatments

These groups of treatments are widely available but very poorly researched in the scientific manner that the medical profession requires to make evidenced-based clinical decisions.
This can make it difficult to fully counsel a woman as to how effective the treatment is, how long any effect will last and, most importantly, how safe the therapy is.
Efficacy is usually limited and of a short duration, with the potential for interactions with other pharmaceutical agents.

Non-hormonal prescription treatments

This group of therapies is increasingly important to consider in the management of women to reduce symptoms of hot flushes when hormones are not wanted or contraindicated, for example previous diagnoses of hormone-sensitive cancers such as breast cancer.

Other issues specific to the menopause can also be treated without hormones using prescribable preparations. These include vaginal moisturizers and lubricants for vaginal dryness and some of the many therapies for treating osteoporosis, which include bisphosphonates, raloxifene denosumab and teriparatide.

Hormones replacement therapy

One of the most effective agents to treat estrogen deficiency symptoms is estrogen.

Treatment should be targeted to the individual woman's needs.

HRT offers the potential for benefits which outweigh harm, providing the appropriate regimen is chosen in terms of dose, route of administration and combination .

Types of HRT

Hormone replacement therapy (HRT) is a treatment that include a fundamental element, the estrogen and progestational agent.

The progestational agents include the natural progesterone and its synthetic analogues (progestins and progestogens). The role of progesterone and its synthetic analogue is limited to the protection of the endometrium only.

In principle, types of regimens prescribed in the treatment of the menopause are:

- estrogen-only regimens
- sequential combined regimens
- continuous combined regimens
- synthetic steroidal alternatives, e.g. Tibolone.

The following diagram shows some of the different HRT preparatio

Sequential Combined Cyclical HRT	**Estrogen daily plus Progesterone/Progestogen for 10-14 days every 28-30 days cycle**	**Short exposure to progesterone/progestogen Cyclical Withdrawal Bleeding**
Sequential Long Cycle HRT	**Estrogen daily for 12 weeks Plus Progesterone/Progestogen added in the last 14 days**	**Higher dose less frequent exposure to progesterone/progestogen Withdrawal Bleeding every 3 months**
Continuous Combined HRT	**Estrogen Plus Progesterone/Progestogen administered together continuously**	**Continuous exposure to progesterone/progestogen No Withdrawal Bleeding**
Continuous progestogen regimen: Tibolone	**Synthetic steroid with estrogenic, androgenic and progestational activity**	**Continuous exposure to progestational effects No Withdrawal Bleeding**

Endometrial protection

For women with intact uterus, protection of the endometrium with progesterone or a progestational agent is required to avoid irregular bleeding, endometrial hyperplasia or even endometrial carcinoma if unopposed estrogen is used.

These agents can be added to estrogen either cyclically for short monthly courses, quarterly, or as a continuous addition.

Estrogen preparations for systemic use

- Estradiol either in a transdermal form (patches and gels) or tablets (micronised or esterified estradiol preparations as in valerate or hemihydrate).
- Conjugated equine estrogens tablets.
- Estradiol given as subcutaneous implants.

Estrogens for vaginal application:

- estradiol cream
- estradiol vaginal tablet
- estradiol vaginal ring.

- **Progestogens:**

-norethisterone;
-levonorgestrel;
-dydrogesterone;
-medroxyprogesterone acetate;
-drospirenone;
-micronized progesterone.

Tibolone

A synthetic steroid, tibolone (known as Livial) was introduced in the treatment of the menopause around 30 years , it is effective in relieving vasomotor symptoms and prevents bone loss. Tibolone is also reported to improve libido, but whether this is a specific androgen receptor agonist effect or secondary to the general improvement in menopausal symptoms is unclear.

Testosterone : given to women with disorders of sexual desire and energy levels who have failed to respond to normal HRT.

HRT routes

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Key benefits of HRT :

- Symptoms improved: vasomotor symptoms; sleep patterns; performance during the day.
- Prevention of osteoporosis: increased bone mineral density; reduced incidence of fragility fractures.
- Lower genital tract: dryness; soreness; dyspareunia;
- CVD: preventative effect if started early in menopause.

Contraindications and potential side-effects

Absolute contraindications:

-suspected pregnancy;

-breast cancer;

-endometrial cancer;

-active liver disease;

-uncontrolled hypertension;

-known current venous thromboembolism (VTE);

-known thrombophilia (e.g. Factor V leiden);

Relative contraindications:

-uninvestigated abnormal bleeding;
-large uterine fibroids;
-past history of benign breast disease;
-unconfirmed personal history or a strong family history of VTE;
-chronic stable liver disease;
-migraine with aura.

Side-effects associated with oestrogen:

breast tenderness or swelling; nausea; leg cramps; headaches.

Side-effects associated with progestogen:

-fluid retention;

-breast tenderness;

-headaches;

-mood swings;

-depression;

-acne.

Risks of Hormone Therapy

Cancer

- Breast cancer is without doubt the cancer that attracts most concern from patients and most attention from the world's media.
- It is important to be aware that recent data with oestradiol HRT suggest that mortality from breast cancer is not increased and that certain types of HRT may promote the growth of pre-existing malignant cells rather than initiate tumours.
- Endometrial cancer and ovarian cancer are not considered significant risks with HRT use Endometrial malignancy risk is largely eliminated if women are given progestogens. Incidence of ovarian cancer has not been shown to significantly increase with HRT use.

Cardiovascular disease and stroke

As discussed above, most of the effects of HRT on the cardiovascular system when given to younger women are beneficial. However, when given to older women the effects may become deleterious.

The degree to which this happens is unclear but is likely to be higher in women taking combined HRT.

Stroke incidence has a similar age effect, with the increased incidence greater in the older woman. The effect is small and is only on the incidence of ischaemic stroke, thought to be an increase of an additional 2 women per 10,000 women per year when on HRT.

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Venous thromboembolism

The influence of HRT on the clotting system is similar to that of the oral contraceptive. The background incidence of all VTE in women over 50 is low (approximately 15–20 per 10,000) and HRT doubles this risk.

There is evidence to suggest that transdermal HRT, through its avoidance of effects on the liver, may not have such a great effect on VTE incidence

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conclusion

- The menopause is defined as the woman's final menstrual period.
- The diagnosis of menopause is a largely clinical diagnosis.
- The hormonal changes during and after the menopause have radical changes on the woman.
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